Marketing ROI & Effectiveness: Analyst Brief

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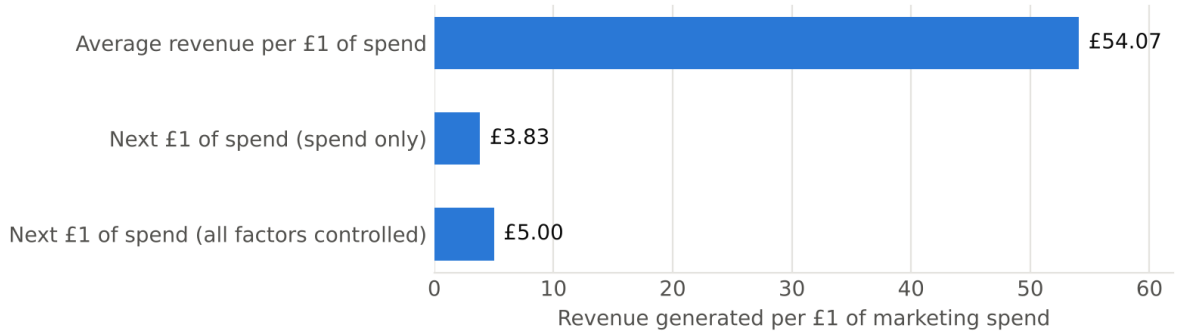
Question

Is marketing spend across four social channels (Facebook, Instagram, TikTok, Twitter) efficient, and where should the next pound go? Analysis of 44,359 cleaned transactions (teaching dataset, 49,287 raw) in R.

Key findings

Finding	Evidence	So what
Average return is high	£54.07 revenue per £1 spend (ROMI 53×)	Headline looks very profitable
Marginal return is far lower	Extra £1 → £3.83–£5.00; elasticity 0.07	Budget on the next pound, not the average
Twitter leads on volume	80% of transactions and revenue; sale value +0.3%	Efficiency edge too small to justify a shift alone
Promotions add no value	No raw difference (p = 0.36); –£22 with controls	Promotion cost needs separate justification
Other segments are minor	Product, buyer, quarter: eta-squared up to 0.03	Not useful budget levers

The next £1 of spend returns about a tenth of the average £1



Method

- **Data validation:** removed negative commissions, missing and invalid category codes; merged duplicate buyer labels; flagged a likely capped time field.
- **KPIs:** portfolio ROMI from totals rather than averaged ratios; volume vs revenue share by channel.
- **Statistics:** ANOVA with effect sizes (eta-squared) and Tukey HSD; multiple regression with all levers (R^2 = 0.53); log-log elasticity; spend-decile analysis.
- **Correction of my first version:** a falling ROMI-vs-spend trend was a by-product of the ROMI formula (r = 0.98 with 1/spend), so I replaced it with a marginal-return test.

Recommendations

- Report marginal ROMI alongside average ROMI in budget reviews.
- Test before reallocating to Twitter (hold-out regions or time periods).
- Review the promotion programme's cost, as it shows no revenue lift.
- Capture product margins and channel costs so ROMI measures profit, not just revenue.

Skills shown

Data cleaning and validation · KPI design · statistical testing with effect sizes · regression · critical review of my own earlier analysis · commercial recommendations · reproducible R code